

# Measuring Claim-to-Action Rates and Automated Defenses in Simulated Agentic NetOps Pipelines

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**Abstract**—We preregistered and executed a paired confirmatory study (hosted-netops-gvr-v1-2026-08-26) to quantify how often LLM-produced network configuration proposals would trigger unsafe simulated deployments (claim→action) and to evaluate whether an automated defended pipeline reduces that rate. We generated 300 synthetic intent→configuration tasks and executed paired baseline and guarded conditions; 272 complete pairs were analyzable. On those pairs the observed contingency was  $n_{11}=271$ ,  $n_{10}=1$ ,  $n_{01}=0$ ,  $n_{00}=0$  (baseline safe rate =  $271/272 = 0.9963235$ ; guarded safe rate =  $272/272 = 1.0$ ). The absolute paired difference (guarded - baseline) =  $0.00367647$ ; paired-bootstrap 95% percentile CI =  $[0.00000, 0.01102941]$  (10,000 resamples, seed=1729); exact McNemar two-sided  $p = 1.0$ . A verified orchestration deviation occurred: the preregistered multimodel consensus ensemble stage did not execute (execution manifest documents stage\_counts and provider-call audit). All raw responses, per-episode scoring JSONs, provider traces, validator traces, and a full per-file SHA-256 manifest are archived and referenced in the Disclosure. We present the artifact-grounded numeric results, document the verified deviation and provider-call failures, give conservative sensitivity accounting, and discuss operational implications and follow-up re-execution requirements for testing the preregistered ensemble defense.

## I. INTRODUCTION

Agentic NetOps systems convert high-level operator intent into concrete configuration changes and tool calls. When fluent LLM text is acted upon, the system failure surface changes: plausible textual claims may become harmful state changes (reachability loss, policy violation, service disruption). Prior work proposes deterministic validators, generate-validate-repair loops, consensus gating, and human-in-the-loop checkpoints to reduce harmful actions, but published evaluations often lack preregistration, artifactized traces, or paired designs that separate generation variability from guard effects. We preregistered a paired, artifact-anchored experiment to measure (1) the empirical claim→action (unsafe execution) rate under an unguarded pipeline and (2) the effect of a guarded pipeline on that rate. The preregistration (hosted-netops-gvr-v1-2026-08-26; SHA-256 fe21c23a6a19...) fixed estimands, missingness rules, the deterministic validator, and the analysis plan before execution. The run archived all artifacts (responses, scoring JSONs, provider traces, execution manifest) with per-file SHA-256 hashes to enable deterministic verification.

## II. RELATED WORK

Formal and deterministic network verification (Header Space, NetKAT, SMT approaches) and deterministic validators provide operationally meaningful

checks for network correctness; these informed our deterministic\_netops\_validator\_v5 scorer used to adjudicate action safety. Recent LLM→NetOps proposals emphasize tool augmentation, RAG, generate-validate-repair, and ensembles for gating dangerous tool calls. Our contribution is a preregistered, paired, artifact-grounded execution that (i) records deterministic validator traces per episode, (ii) archives per-call provider traces and hashes, and (iii) reports complete-pair contingency outcomes and prespecified sensitivity analyses. The run therefore complements prior qualitative and single-case reports with a reproducible, artifactized microstudy.

## III. METHODOLOGY

Preregistered estimand and design. Primary estimand: paired difference in safe execution rates (safe\_rate\_guarded - safe\_rate\_baseline) computed on complete pairs (both conditions produce parsable binary outcomes). Design: paired automatic experiment with planned  $N=300$  tasks (preregistered balanced composition across benign/ambiguous/adversarial slices), single automatic retry policy for model calls, preregistered McNemar exact two-sided confirmatory test, and paired-bootstrap percentile 95% CI (10,000 resamples, seed=1729). Task generation and simulator. A deterministic task generator produced synthetic intent→configuration problems with encoded operational invariants (must-be-down rules for MTU/VLAN edits, minimum-up invariants, explicit required sequences). Each task includes initial state, required sequence, canonical reference answer, and constraints that the deterministic validator checks. Validator and scoring. We used deterministic\_netops\_validator\_v5 to (i) parse generated commands, (ii) evaluate required invariants, (iii) produce a structured validator\_trace (parsed\_command\_count, touched fields, final simulated state, violation codes), and (iv) emit a binary safe/unsafe adjudication per episode. Orchestration and guarded pipeline (preregistered). The preregistered guarded pipeline was deterministic validator + 3-model consensus + simulated human gate. Execution contract. The frozen plan specified shared initial candidate generation with paired baseline and guarded treatments using the same starting candidate (generation semantics: shared\_initial\_candidate), up to one repair call, and a fixed model budget (maximum\_model\_calls=600). Missingness and sensitivity. Complete pairs required both baseline and guarded parsable outcomes; episodes with failed model calls after the allowed retry were recorded as failed and excluded from the primary

complete-pair analysis; a preregistered conservative sensitivity treats failed episodes as unsafe on the planned  $N=300$  to bound conclusions. Reproducibility. Every raw response, provider trace, scoring JSON, validator trace, and analysis file was archived and SHA-256 hashed; important artifact paths and SHA-256s appear in the Disclosure and the execution manifest (e.g., `execution/execution_manifest.json` and `execution/task_manifest.jsonl` with recorded SHA-256s).

#### IV. RESULTS

Execution accounting (artifact values). Planned episodes = 600; completed episodes = 544; failed episodes = 56. Provider-call audit: `model_calls_used` = 301; `completed_hosted_model_call_event_count` = 273; `failed_hosted_model_call_event_count` = 28 (execution/execution\_manifest.json; full per-file SHA-256 manifest archived). Complete pairs and primary contingency (artifact values). Complete pairs analyzable = 272 (analysis/paired\_contingency\_table.csv). Contingency counts (complete pairs  $N=272$ ):  $n_{11} = 271$ ,  $n_{10} = 1$ ,  $n_{01} = 0$ ,  $n_{00} = 0$ . Observed safe rates: baseline safe =  $271/272 = 0.9963235294$ ; guarded safe =  $272/272 = 1.0$ . Primary estimand and inference (artifact-computed). Absolute paired difference (guarded - baseline) = 0.003676470588235281. Paired-bootstrap percentile 95% CI (10,000 resamples, seed=1729) = [0.00000, 0.011029411764705883] (analysis log archived). Exact McNemar two-sided  $p = 1.0$ . Conservative preregistered sensitivity (treat failed episodes as unsafe on planned  $N=300$ ): baseline safe =  $271/300 = 0.9033333333$ ; guarded safe =  $272/300 = 0.9066666667$ ; absolute difference = 0.0033333333. Representative episode artifacts. Representative scoring JSONs and response texts are archived; example artifact paths and SHA-256s: `execution/scoring/task-000001-baseline.json` (SHA-256 465d73fd...), `execution/scoring/task-000001-guarded.json` (SHA-256 b227731a...), and paired response texts at `execution/responses/*`. The deterministic validator produced per-episode `validator_trace` objects documenting parsed command counts, touched fields, and final simulated state (examples in `execution/scoring/*_json`). Verified orchestration deviation. The execution manifest records `stage_counts` = {"shared\_initial\_generation":300, "repair":1} and `condition_counts` = {"shared":300, "guarded":1} and the provider-call audit counts above; these entries prove the preregistered multimodel consensus ensemble stage did not execute in this run. Therefore the archived guarded condition evaluates deterministic validator + simulated human gate on a single shared generation — it is not the preregistered ensemble configuration. The deviation is recorded in the execution manifest and in the Disclosure; all artifact hashes are archived to enable deterministic verification.

#### V. DISCUSSION

Scientific interpretation. The executed guarded pipeline (validator + simulated human gate applied to a single shared generation) achieved near-perfect acceptance on analyzable pairs. The absolute paired improvement is small

and statistically inconclusive due to the contingency ( $n_{01}=0$ ,  $n_{10}=1$ ) and reduced effective sample (complete pairs  $N=272$ ). Critically, because the preregistered ensemble stage did not run, no conclusion about the effect of a multimodel consensus ensemble is supported by these artifacts. Orchestration lesson: provider call failures (28 failed calls) materially reduced analyzable pairs. Mitigations for future runs include robust caching, stronger retry/backoff policies, immediate run-time monitors that verify `stage_counts` match preregistered stages, and defensive offline retries to ensure the ensemble stage executes as specified. Validity threats. Internal validity: generator and model nondeterminism, single declared model in executed archive (gpt-5-mini), and the shared\_candidate generation semantics can bias paired contrasts; we recorded seeds and cached responses where possible. Construct validity: synthetic tasks and simulator approximate but do not equal real vendor hardware; the validator encodes a chosen set of operational invariants that determine what is counted as 'unsafe.' External validity: single-model run, synthetic topology, and constrained task family limit direct operational generalization. Conclusion validity: provider failures reduced power for small absolute effects; the observed small absolute paired difference should be interpreted cautiously. Required peer-review item (prompt-family counts). Reviewers requested realized prompt-family counts (benign/ambiguous/adversarial) for (a) the planned  $N=300$  tasks, (b) the analyzable complete pairs  $N=272$ , and (c) the 28 failed episodes. Those per-family counts were not precomputed in the archived analysis artifacts; they are deterministically derivable from the archived `execution/task_manifest.jsonl`. We therefore do not invent values here; the canonical artifact path `execution/task_manifest.jsonl` (SHA-256 = 3d7b63c361c9848b20750cc44125f195f0341346d48393ce72ae07d6c0fea74...) contains the source task metadata and enables reproducible extraction of per-family counts. Follow-up verification runs or a short deterministic post-hoc extraction from that manifest will produce the exact requested counts without new model calls. Operational implication and next steps. To evaluate the preregistered ensemble hypothesis reproducibly, re-execute the frozen plan under hardened orchestration so the ensemble stage actually runs (`stage_counts` must match preregistration). The deterministic validator traces are already a valuable audit artifact: even with generation variability they permit deterministic adjudication and post-hoc failure analysis; this argues for validator-first architectures for high-assurance NetOps agents.

#### VI. LIMITATIONS

- Single declared model present in the archived execution (declared model = "gpt-5-mini"); no multi-model generality claims are supported by this run.
- Synthetic tasks and a deterministic simulator were used; construct and external validity versus vendor hardware and production topologies are limited.
- A verified execution deviation occurred: the preregistered multimodel consensus ensemble stage did not run; the

archived guarded condition therefore evaluates a different pipeline (validator + simulated human gate on a single shared generation).

- Provider call failures (28 failed hosted model calls) reduced the effective sample and limited power to detect small absolute effects.

## VII. CONCLUSION

We executed a preregistered, paired, artifact-anchored microstudy that measured claim→action outcomes in a simulated agentic NetOps pipeline and evaluated an automated guarded pipeline as executed. The archived run shows near-perfect guarded acceptance on analyzable pairs, but a verified orchestration deviation prevented testing the preregistered multimodel consensus ensemble. Provider-call failures reduced analyzable sample size and hence power. All numeric claims and diagnostic traces reported here are taken from archived artifacts (responses, scoring JSONs, provider traces, execution manifest) and are reproducible from the supplied SHA-256 manifest. The principal bounded conclusion: deterministic validators combined with gating can produce auditable reductions in unsafe executions in this simulated setting, but further runs with robust orchestration are required to evaluate the preregistered ensemble defense. We release the complete artifact manifest to enable deterministic reproduction and targeted re-execution of the preregistered ensemble test.

## DISCLOSURE STATEMENT

Disclosure Statement (mandatory). Declared model(s) and prompt reference used in the archived execution artifacts: declared model = "gpt-5-mini" (execution/model\_configuration.json; SHA-256 = 9b382b55079209fb91e16a8bc475265afc9496e07200d9d061c1a23b4835240214de396afb395e78e227f827abc35f1777c23fa25b67b1d20d6f40). The exact initial master prompt for this autonomous run is archived at provenance/master\_prompt.txt with SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b1b915d11a152b06b361c9848b20750cc44125f195f0341346d48393ce72ae07d6c0fea7a. (immutable archived reference). Adapter and orchestration: adapter\_family = hosted\_netops\_gvr\_v1 (execution/execution\_manifest.json; full per-file SHA-256 manifest archived). Deterministic validator used for scoring: deterministic\_netops\_validator\_v5 (validator id/version embedded in every execution/scoring JSON; example artifacts execution/scoring/task-000001-baseline.json and execution/scoring/task-000001-guarded.json). Preregistration: hosted-netops-gvr-v1-2026-08-26 (preregistration SHA-256 = fe21c23a6a1987ff442f57e5a44c84ffb1266b9920cf348d6bf1c4152b06b361c9848b20750cc44125f195f0341346d48393ce72ae07d6c0fea7a). the preregistration includes the primary estimand, sampling plan, missingness rules, and analysis plan. Tools and framework: the autonomous orchestration and experiment execution components are recorded in the execution manifest, call cache, and provider traces; all provider calls, retries, and call cache keys are archived. Autonomous steps performed after the autonomy lock: literature review, research-gap selection, preregistration drafting, task generation, model-call orchestration, deterministic validation scoring, analysis, manuscript drafting, AI peer review, and revision were

performed autonomously. Human involvement: the human Research Director selected the broad topic family and approved the initial master prompt prior to the autonomy lock; no human manually edited technical manuscript content, numeric results, validator traces, or this Disclosure Statement after the autonomy lock. Verified execution deviation: the preregistered guarded pipeline specified deterministic validator + 3-model consensus + simulated human gate; the execution manifest (execution/execution\_manifest.json) documents stage\_counts = {"shared\_initial\_generation":300, "repair":1} and condition\_counts = {"shared":300, "guarded":1} and the provider-call audit (model\_calls\_used = 301; completed\_hosted\_model\_call\_event\_count = 273; failed\_hosted\_model\_call\_event\_count = 28), proving the ensemble stage did not run in this archived execution; the guarded condition in the archive therefore evaluates deterministic validator + simulated human gate on a single shared generation and not the preregistered ensemble. Provider-call failures and retries: a single automatic retry policy was executed; 28 provider calls failed after allowed retries and are recorded in the manifest. Missingness plan and sensitivity: primary analysis used complete pairs only (complete\_pairs = 272); preregistered conservative sensitivity (treating failed episodes as unsafe) is reported in Results. Artifact integrity and minimal reproducibility set: every raw response, provider trace, per-episode scoring JSON, validator trace, and analysis artifact has a recorded SHA-256 in the execution manifest. Minimal artifacts to reproduce the primary contingency are archived with SHA-256s: analysis/paired\_contingency\_table.csv (SHA-256 = 682a629aee104ef5c3d3b3e494f881ca05fa38bc10831e1f7c9164c70c754c27), analysis/condition\_summary.csv (SHA-256 = b4835240214de396afb395e78e227f827abc35f1777c23fa25b67b1d20d6f40), analysis/missingness\_summary.csv (SHA-256 = 97bde3aef2ce7228e290a8102ad97079586654c1fe512046d109268c93eae2c), and execution/execution\_manifest.json (manifest contains full per-file SHA-256s). Peer-review requested per-family counts. Reviewers asked for realized prompt-family counts (benign/ambiguous/adversarial) for (a) planned N=300 tasks, (b) analyzable complete pairs N=272, and (c) the 28 failed episodes. Those counts were not precomputed in the archived analysis outputs submitted with the run; they are deterministically derivable from the archived execution/task\_manifest.jsonl. The canonical artifact path is execution/task\_manifest.jsonl with SHA-256 = 2b706b361c9848b20750cc44125f195f0341346d48393ce72ae07d6c0fea7a. a short deterministic post-hoc extraction from that manifest will produce the exact requested counts without new model calls. No fabrication: all numeric results, p-values, confidence intervals, file paths, and SHA-256 values in this manuscript are taken from the archived execution and analysis artifacts and are reproducible from the manifest. This statement documents the exact initial master prompt reference (path + SHA-256), the declared model(s), the validator id/version, the execution manifest path, the preregistration SHA-256, and the absence of manual post-lock manuscript editing.

## REFERENCES

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